# Staggered Modules in Logarithmic CFT

## From Free Fields to Liouville Theory

YUN-YEN, WU

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# Outline

- Introduction to Logarithmic Conformal Field Theory
- Free Field Realizations
- Liouville Theory and W Matrix
- Conclusion & Future Work

# Introduction to Logarithmic Conformal Field Theory

# 1+1D Conformal Field Theory

- Local field $\phi(z,\bar{z})$ lives on the complex plane (radial quantization)
- Conformal symmetry → Infinite-dimensional algebra (holomorphic + antiholomorphic)
- Primary fields transform covariantly

$$\phi'(w,\bar{w}) = \left(\frac{dw}{dz}\right)^{h}\left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}}\phi(z,\bar{z})$$

- Chiral energy-momentum tensor

$$T(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}$$

- Virasoro algebra with central charge

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}$$

- Hilbert space decomposes into irreducible representations $\mathcal{H} = \bigoplus_i \mathcal{V}_{h_i} \otimes \bar{\mathcal{V}}_{\bar{h}_i}$

# Highest Weight Representations

- Highest weight primary state satisfies

$$L_0|h\rangle = h|h\rangle \quad \& \quad L_n|h\rangle = 0 \quad (n > 0)$$

- Verma Module (generated by lowering operators)

$$\mathcal{V}_h = U(\mathrm{Vir}_{<0})|h\rangle$$

- Diagonalizable property of $L_0$ at every module

- Two-point: $\langle \phi_h(z)\phi_h(0)\rangle = \frac{\mathrm{A}}{z^{2h}}$
- Three-point: $\langle \phi_1\phi_2\phi_3\rangle = \frac{C_{123}}{z_{12}^{h_1+h_2-h_3} z_{23}^{h_2+h_3-h_1} z_{12}^{h_1+h_3-h_2}}$

Conformal Invariance!

# Singular and Null Vector

- Singular Vector: generated by a specific combination of creation operator $U_{\Delta h}$ acting on the vacuum $h$

$$w = U_N v_L = \sum c_{\{n\}} L_{-n_1} \dots L_{-n_k} |h\rangle \quad \text{with} \quad \sum n_i = N$$

- Annihilated by all positive Virasoro modes and the conformal dimension $h_w = h + N$ is specific (determined by its level of descendant $N$)

- Null Vector / Null Condition has zero norm and decouples from correlators

$$\langle w|w\rangle = 0$$

# Reducible vs. Irreducible Virasoro Modules

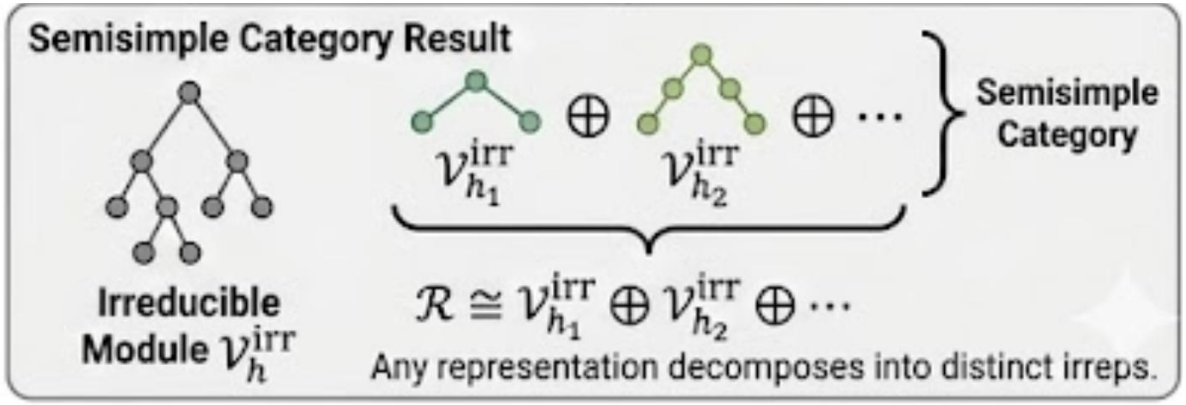
- A Verma module $\mathcal{V}_h$ is **reducible** if contains a singular (null) vector:

$$|\chi\rangle = U(L_{-n})|h\rangle \quad \text{with} \quad L_m|\chi\rangle = 0\ \forall m > 0$$

- Irreducible module = quotient by the null submodule $\mathcal{U}$

$$\mathcal{V}_h^{\text{irr}} = \mathcal{V}_h/\mathcal{U}$$

- Semisimple representation category (direct sums of irreps)



# Definition of Logarithmic CFT

- Logarithmic CFTs are the natural generalization in which the representation category is not semisimple: the state space $\mathcal{H}$ contains indecomposable but reducible Virasoro modules ($L_0$, $\bar{L}_0$ is not diagonalizable)
- Rank-2 Jordan block

$$\hat{L}_0|\phi\rangle = h|\phi\rangle, \qquad \hat{L}_0|\psi\rangle = h|\psi\rangle + |\phi\rangle, \qquad \hat{L}_{n>0}|\psi\rangle = 0 \qquad \hat{L}_0 = \begin{pmatrix} h & 1 \\ 0 & h \end{pmatrix}$$

primary field  logarithmic partner

- Two-point function (in 1+1D)
  - $\langle\phi(z)\phi(0)\rangle = 0$ zero norm for primaries
  - $\langle\phi(z)\psi(0)\rangle = \frac{A}{z^{2h}}$
  - $\langle\psi(z)\psi(0)\rangle = \frac{B-2A\log z}{z^{2h}}$ "A" is the logarithmic coupling (zero in standard CFT)

A basis change do not alter the Jordan structure. No free parameters.

# Staggered Module

- A staggered module $M$ is an indecomposable Virasoro module that fits into a non-split short exact sequence

$$0 \longrightarrow H_L \xrightarrow{\iota} M \xrightarrow{\pi} H_R \longrightarrow 0 \quad (H_L \text{ and } H_R\text{: highest}-\text{weight modules})$$

- Exactness means: glue them together!
- - ι: $H_L \to M$ is injective
- - π: $M \to H_R$ is surjective
- - $M/\iota(H_L) \cong H_R$
- Non-split: $M \neq H_L \oplus H_R$ as Virasoro modules

# Chiral vs. Diagonal/Non-Chiral

- Chiral Logarithmic Module: a representation of a single Virasoro algebra

$$\text{Ex:} \quad 0 \to X_1 \to M \to X_2 \to 0 \quad \text{only left\textbackslash right}-\text{moving sector involved}$$

- Purely algebraic object, logarithms appear already in chiral correlators
- Diagonal (Non-Chiral / Bulk) Logarithmic Structure

$$\mathcal{H} = \bigoplus_{i,j} M_{ij} \underset{\text{chiral}}{\mathcal{V}_i} \otimes \underset{\text{anti-chiral}}{\bar{\mathcal{V}}_j}$$

- Jordan blocks appear in left, right or diagonally

$$(L_0 + \bar{L}_0)|\Psi\rangle = \Delta|\Psi\rangle + |\Phi\rangle$$

# Indecomposable Representations

- Set a normalized highest-weight vectors $v_L \in H_L$ and $v_R \in M$ be projected to the highest-weight vector $H_R$
- Unique (up to scalar) homogeneous operator $U_{\Delta h} \in U(\mathrm{Vir}_{<0})$ such that the connecting relation holds

$$(L_0 - h_R)v_R = U_{\Delta h} v_L$$

- Indecomposable parameter (logarithmic coupling) is the gauge-invariant scalar

$$\beta(M) := \langle v_L, (U_{\Delta h})^\dagger v_R \rangle$$

Contravariant Shapovalov form

$$U_{\Delta h}^\dagger v_R = \beta(M) v_L$$

- $\beta(M) \neq 0$ implies $M$ is staggered and indecomposable
- $\beta(M)$ is independent of the choice of $v_R \to v_R + \iota(\eta)$ and of the representative of $U_{\Delta h} = L_{-1}^{\Delta h} + \cdots$

( $\eta \in H_L$ is the vector at level $\Delta h$ )

# Why Staggered Modules Appear

- Ordinary CFT on a torus (reducible $\rightarrow \beta(M) = 0$)

$$Z(\tau) = \sum_h |\chi_h(\tau)|^2 \quad \text{satisfy} \quad Z(-1/\tau) = Z(\tau)$$

holomorphic func

- For certain non-unitary theory (e.g., symplectic fermion at $c = -2$ )
- The ordinary sum is not invariant under $\tau \rightarrow -1/\tau$ and break modular invariance
- Modular invariance is restored by including indecomposable (staggered) modules in the partition sum

$$\chi_{\log}(\tau) \sim \frac{\log q}{2\pi i} \chi_{\text{primary}}(\tau)$$

# Relation to Standard CFT & Examples

- Standard CFT: all Virasoro modules are completely reducible $\rightarrow$ $\beta = 0$
- LCFT (e.g. percolation $c = 0$, symplectic fermion $c = -2$) appear as indecomposable representations required by modular invariance
- Concrete examples (Virasoro, $c = 0$):

$$0 \rightarrow V_0/V_3 \rightarrow S_{1,5} \rightarrow V_0/V_1 \rightarrow 0 \quad \text{with} \quad \beta_{1,5} = -5/18$$

$$0 \rightarrow V_0/V_2 \rightarrow S_{1.4} \rightarrow V_1/V_5 \rightarrow 0 \quad \text{with} \quad \beta_{1,4} = -1/2$$

# Free Field Realization

# Free Field Construction

- Bosonic Fock space $\mathcal{F}_\eta$ ( Heisenberg algebra)

$$\mathcal{F}_\eta = U(\hbar) \otimes |\eta\rangle$$

$$[\alpha_m, \alpha_n] = m\delta_{m,-n}\mathbf{1} \quad \text{with} \quad [\alpha_m, \mathbf{1}] = 0$$

- Free-field energy momentum tensor

$$T(z) = \frac{1}{2} : \partial a(z)^2 : + \lambda \partial^2 a(z)$$

- Action of $\mathcal{L}_n$ and conformal dimension

$$L_n = \frac{1}{2} \sum_{k \in \mathbb{Z}} : a_k a_{n-k} : - \lambda(n+1) a_n$$

$$h_\eta = \frac{1}{2} \eta(\eta - 2\lambda)$$

# Extended Kac Table

- Reducible $(p, q)$ models: for coprime $p, q > 0$

$$c_{p,q} = 1 - \frac{6(q-p)^2}{pq}$$

$$h_{r,s} = \frac{(rq - sp)^2 - (q-p)^2}{4pq}, \qquad r, s \in \mathbb{Z}^+$$

- Two Fock spaces

$$\lambda^{\pm}_{p,q} = \pm \frac{q-p}{\sqrt{2pq}} \quad \text{and} \quad \eta^{\pm}_{r,s} = \frac{1}{\sqrt{2pq}}[(q-p) \pm (rq - sp)]$$

1. Reducibility: $F_{\eta^{\pm}}$ has singular vectors exactly at $(c, h_{r,s})$
2. Integrality: $\Delta h = h_{\eta_L} - h_{\eta_R}$ must be an integer

# Free-Field Vertex Operators

- Vertex operator of momentum $\mu$

$$V_\mu(z) =: exp\big(\mu a(z)\big): = \sum_{n\in\mathbb{Z}} V_n z^{-n-h_\mu}$$

Zero mode $\oint_{z=0} z^{h_\mu - 1} V_\mu(z) dz = V_0 : \mathcal{F}_\eta \to F_{\eta+\mu}$ when $\lambda = \frac{1}{2}\mu - \frac{1}{\mu}$

$$[L_m, V_0] = 0 \text{ for all } m$$

# Staggered Fock Module

- Rank-2 Vir-module $M$ ($M \simeq F_L \oplus F_R$ as vector space)

$$0 \to F_L \to M \to F_R \to 0$$

$$L_n = \begin{pmatrix} L_n & V_n \\ 0 & L_n \end{pmatrix} \quad \text{on } F_L \oplus F_R,$$

- How to construct the $V_n$?

1. A Vir-intertwiner $V_0 : F_R \to F_L$

$$V_0|\eta_R\rangle \propto U_{\Delta h}|\eta_L\rangle$$

1. Define

$$V_n := \frac{1}{N\mu}[\alpha_n, V_0]$$

$\mu$: momentum parameter
$N$: number of vertex operators

- Indecomposable parameter

$$\beta = \frac{1}{\Delta h N \mu^2} \text{(sum over partitions with Selberg–integral coefficients )}$$

# Liouville Theory and W Matrix

# Fermionic Setup & Bosonization

- Dirac fermion with non-Hermitian mass term $L = \text{Dirac Term} + \Delta\psi_L^\dagger\psi_R$
- Bosonization Result $L = \frac{1}{4\pi}\left(\partial_\mu\phi\right)^2 + \Delta e^{2b\phi}$ with $b = \frac{i}{\sqrt{2}}$ and $c = 1 - 6\left(b + \frac{1}{b}\right)^2 = -2$

  compactified imaginary Liouville theory

- The Hermitian Baseline (Projected onto even fermion parity state)
- NS-NS sector (anti-periodic): Map to an untwisted local boson field with $\left(h, \bar{h}\right) = (0,0)$
- R-R (periodic) sector: derived from twisting the NS-NS sector; map to an untwisted local boson field with $\left(h, \bar{h}\right) = \left(\frac{1}{8}, \frac{1}{8}\right)$
- Locality shift on periodic part (R-R sector) due to the twist

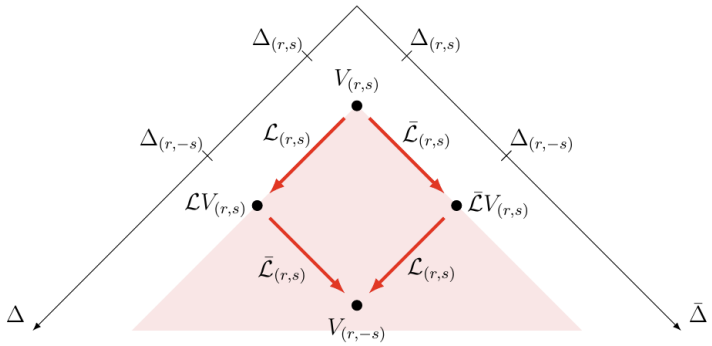
# The Logarithmic Field

- Consider $c = -2$ case
- Introduce the linear combination as follows

$$W^{\kappa}_{r,s} = (1-\kappa)V'_{r,-s} + \kappa L\bar{L}V'_{r,s}$$

- Algebraic definition

$$(L_0 - \Delta_{r,-s})W^{\kappa}_{r,s} = (\bar{L}_0 - \Delta_{r,-s})W^{\kappa}_{r,s} = V_{r,-s}$$




primaries of different dimensions

- $\kappa$ is the only free parameter and normalization-independent

$$L_{r,s}DW^{\kappa}_{r,s} = \kappa\left(L_0 - \Delta_{r,-s}\right)W^{\kappa}_{r,s}$$

$D$: annihilation operator of degree $rs$

$$L_0\begin{pmatrix} W^{\kappa}_{(r,s)} \\ V_{(r,-s)} \end{pmatrix} = \begin{pmatrix} \Delta_{(r,-s)} & 1 \\ 0 & \Delta_{(r,-s)} \end{pmatrix}\begin{pmatrix} W^{\kappa}_{(r,s)} \\ V_{(r,-s)} \end{pmatrix}$$

# Mutual Locality

- Bosonic vertex operators $V_{m,w}$ are defined by their momentum $m$ and winding $w$
- Local algebra – local fermions naturally satisfy the parity rule $m \equiv w \pmod 2$

→ even-even and odd-odd sublattices

- Non-local algebra – still exist and act as twist operators

→ even-odd and odd-even states

- Ex: Romand spin field $S^{\pm}$ with $(m, w) = (1,0)$
- non-local w.r.t. fermions and generate the branch cuts twist the boundary condition ($NS \leftrightarrow R$)

# Liouville Field Theory

- Action

background charge coupling
(fix conformal anomoly)

$$A_L[\hat{g},\varphi] = \frac{1}{4\pi}\int \sqrt{\hat{g}}\left[\hat{g}^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi + Q\hat{R}\varphi + 4\pi\mu e^{2b\varphi}\right]d^2x$$

- Central charge and conformal dimension in terms of coupling constant and momentum

$$c = 1 - 6\left(\beta - \frac{1}{\beta}\right)^2 \quad , \quad \Delta_{(r,s)} = \frac{c-1}{24} + P_{(r,s)}{}^2 \qquad \beta = ib$$

- Null vector condition

$$\mathcal{V}_\Delta \text{ has a null vector} \quad \longleftrightarrow \quad \Delta \in \Delta_{(r,s)}{}_{r,s\in\mathbb{N}*}$$

- Discrete momentums

the only null vectors are at level $rs$

$$P_{(r,s)} = \frac{1}{2}(\beta r - \beta^{-1}s)$$

# Two Flavors of Liouville Theory

- DOZZ-Liouville theory: $c \in \mathbb{C} - (-\infty, 1)$
- Null descendants vanish: $LV_{r,s} = \bar{L}V_{r,s} = 0$
- $V_{r,s}$ is a degenerate field
- Primary of opposite momentum obtained as derivative

$$L\bar{L}V'_{r,s} = V_{r,-s}$$

- Imaginary / Timelike Liouville theory: $c \leq 1$
- Null descendant do not vanish
- Relation between primaries

$$L\bar{L}V_{r,s} = V_{r,-s}$$

# Logarithmic Rep from Derivatives

- How to get the logarithmic partners? Take the derivative w.r.t. conformal dimension $\Delta$
- Vector of derivatives (n-th order)

$$\hat{V}_\Delta = \left(\frac{1}{n!}V_\Delta^{(n)} \ \dots \ V'_\Delta \ V_\Delta\right)^T$$

- Action of dilation generator

$$\hat{L}_0\hat{V}_\Delta = \hat{\Delta}\hat{V}_\Delta$$

$$\hat{\Delta} = \begin{pmatrix} \Delta & 1 & & \\ & \Delta & 1 & \\ & & \ddots & 1 \\ & & & \Delta \end{pmatrix}$$

Jordan block of size n+1
Non-diagonalizable

# The Role of Twist Operator

$\pm$: fermion number parity
$(-1)^F$ eigenvalue

- Twist Operator in the unperturbed limit ($\Delta = 0$)
- Periodic fermions have integer modes including chiral zero modes obey $\{\psi_0, \psi_0^\dagger\} = 1$
- Acting on a ground-state doublet $\{|R,+\rangle_R, |R,-\rangle_R\}$ gives a conformal dimension of $h_R = \frac{1}{8}$
- The vacua are generated from the NS vacuum by the spin fields

$$|R,\pm\rangle_R = \lim_{z\to 0} S_R^{\pm}(z)|0\rangle_R$$
$$|R,\pm\rangle_L = \lim_{\bar{z}\to 0} \bar{S}_R^{\pm}(\bar{z})|0\rangle_L$$

- The full (non-chiral) Ramond ground states are $|R,\pm\rangle_R \otimes |R,\pm\rangle_L$
- Turning on the non-Hermitian interaction will drastically alter the ground-state structure and the conformal dimensions

# The $\kappa$ Condition

- Why can't we just take the derivative?
- Null condition does not exist here. Taking derivative cannot satisfy the closed Ward identities.

- What condition does $\kappa$ satisfy?
- Fixed by the existence of degenerate field via OPE pole cancellation

$$V_{\langle 1,s_0\rangle}V_{P_{r,0}+\epsilon} \sim C_1(\epsilon)\left(1+f(\epsilon)\mathcal{L}_{r,s}+\cdots\right)V_{(r,s)+\epsilon}+C_2(\epsilon)V_{(r,-s)+\epsilon}+\cdots$$

- $\kappa$ form

$$\kappa_{r,s}^{-}=\frac{P_{r,s}}{P_{r,s}-P_{r,-s}}=\frac{s-r\beta^2}{2s}$$

# Two-point Function

- Two-point functions between logarithmic and primary field

$$\begin{bmatrix} \left\langle W^{\kappa}_{(r,s)} W^{\kappa}_{(r,s)} \right\rangle & \left\langle W^{\kappa}_{(r,s)} V_{(r,-s)} \right\rangle & \left\langle W^{\kappa}_{(r,s)} \bar{\mathcal{L}} V_{(r,s)} \right\rangle \\ \left\langle V_{(r,-s)} W^{\kappa}_{(r,s)} \right\rangle & \left\langle V_{(r,-s)} V_{(r,-s)} \right\rangle & \left\langle V_{(r,-s)} \bar{\mathcal{L}} V_{(r,s)} \right\rangle \\ \left\langle \bar{\mathcal{L}} V_{(r,s)} W^{\kappa}_{(r,s)} \right\rangle & \left\langle \bar{\mathcal{L}} V_{(r,s)} V_{(r,-s)} \right\rangle & \left\langle \bar{\mathcal{L}} V_{(r,s)} \bar{\mathcal{L}} V_{(r,s)} \right\rangle \end{bmatrix}$$

$$= \frac{1}{|z_{12}|^{4\Delta_{(r,-s)}}} \begin{bmatrix} b_1 - \log|z_{12}|^4 & 1 & \frac{2\omega_{(r,s)}}{\rho_{(r,s)}} z_{12}^{rs} \\ 1 & 0 & 0 \\ \frac{2\omega_{(r,s)}}{\rho_{(r,s)}} z_{21}^{rs} & 0 & \frac{2}{\kappa\rho_{(r,s)}} z_{12}^{rs} z_{21}^{rs} \end{bmatrix}$$

- Non-vanishing quantity: $\rho$ and $\omega$

$$\rho_{r,s} = \frac{\partial}{\partial\Delta} (\mathcal{L} V_\Delta L V_\Delta) \Big|_{\Delta_{r,s}}$$ Pure normalization prefactor

$$\omega_{r,s} = \left[L_1^{rs}, L_{r,s}\right](\Delta_{r,s})/(rs)!$$ Fix off-diagonal entries

# Logarithmic Coupling

- Definition of Logarithmic coupling

$$\gamma = \text{the coefficient of the log term in two-point function}$$

- Our result (Take $c = -2$)

$$\gamma = \rho_{r,s}\kappa$$

$$\gamma_{r,s} = \rho_{r,s}\kappa = \beta_M$$

Non-Hermitian free-fermion result

| Index $(r,s)$ | $\rho_{(r,s)}$ | $\kappa^-_{(r,s)}$ | $\gamma^-_{(r,s)}$ | $\Delta_{r,s}$ | $\Delta_{r,-s}$ |
|---|---|---|---|---|---|
| (1,1) | 2 | $-0.5$ | $-1$ | 0 | 1 |
| (2,1) | 12 | $-1.5$ | $-18$ | 1 | 3 |
| (3,1) | 1,080 | $-2.5$ | $-2,700$ | 3 | 6 |
| (4,1) | 453,600 | $-3.5$ | $-1,587,600$ | 6 | 10 |
| (5,1) | 571,536,000 | $-4.5$ | $-2,571,912,000$ | 10 | 15 |

| $M$ | $\beta_M$ |
|---|---|
| 1 | $-1$ |
| 2 | $-18$ |
| 3 | $-2700$ |
| 4 | $-1587600$ |
| 5 | $-2571912000$ |

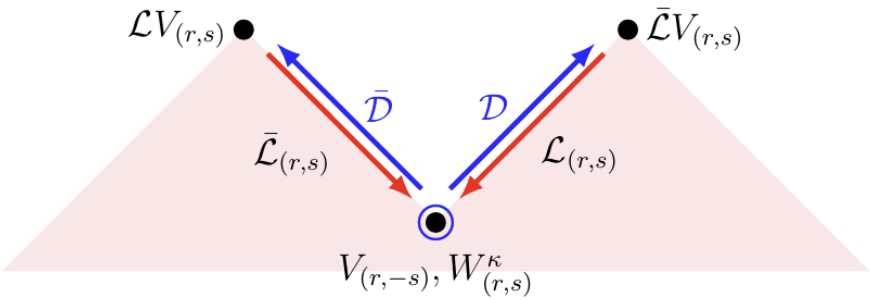
# Staggered Module

- Staggered Module at $c = -2$ ( size 2 )

$$\begin{tikzcd}
 & W_{r,s}^{\kappa} \arrow[dl, "\gamma_{r,s}{}^{-1}(L\bar{L})^{\dagger}"'] \arrow[dd, "L_0 - \Delta_{r,-s}"] \\
V_{r,s} \arrow[dr, "L\bar{L}"'] & \\
 & V_{r,-s}
\end{tikzcd}$$

Indecomposable but reducible

- Two submodules
- $\mathcal{U}_{r,s}$: submodule generated by $\mathcal{L}V_{r,s}$
- $\bar{\mathcal{U}}_{r,s}$: submodule generated by $\bar{\mathcal{L}}V_{r,s}$
- $\mathcal{U}_{r,s} \simeq V_{\Delta_{r,-s}} \otimes \bar{V}_{\Delta_{r,s}}$
- $\bar{\mathcal{U}}_{r,s} \simeq V_{\Delta_{r,s}} \otimes \bar{V}_{\Delta_{r,-s}}$
- $\mathcal{U}_{r,s} \cap \bar{\mathcal{U}}_{r,s} \simeq V_{\Delta_{r,-s}} \otimes \bar{V}_{\Delta_{r,-s}}$



# Non-diagonal Field

- In the bulk (non-chiral) theory we work with full Virasoro × anti-Virasoro modules
- Fields are classified by the conformal weights and spins (left and right momenta differ)

$$S = \Delta - \bar{\Delta} = P^2 - \bar{P}^2 \neq 0$$

Diagonal: scalar
Non-diagonal: spin

- Jordan block structure in the total dilation operator

$$(L_0 + \bar{L}_0)\begin{pmatrix}\Psi\\ \Phi\end{pmatrix} = \begin{pmatrix}\Delta + \bar{\Delta} & 1\\ 0 & \Delta + \bar{\Delta}\end{pmatrix}\begin{pmatrix}\Psi\\ \Phi\end{pmatrix}$$

The logarithmic partner Φ can now live in a different spin sector

- Why non-diagonal fields are needed

1. Crossing symmetry and modular invariance of the full partition function require both sectors
2. Logarithmic extension (operators) must include them to keep the theory consistent

# Conclusion & Future Work

- Conclusion

1. Staggered modules are the fundamental indecomposable but reducible Virasoro representations of logarithmic CFT, fully characterized by the indecomposable parameter $\beta$
2. Free field realization provides a concrete, algebraic framework to construct staggered modules
3. Diagonal Liouville realization via $W^{\kappa}$ fields ($\kappa$ fixed by degenerate OPE)

- Future Work

1. Construct the non-diagonal (non-zero spin) sector of Liouville theory
2. Check the Hilbert space of compactified imaginary Liouville theory

# Reference


- Nivesvivat, R., & Ribault, S. (2021). Logarithmic CFT at generic central charge: from Liouville theory to the Q-state Potts model. *SciPost Physics*.
- Cromer, M. (2016). Free field realisations of staggered modules in 2D logarithmic CFTs. *arXiv:1612.02909*.
- Kytölä, K., & Ridout, D. (2009). On staggered indecomposable Virasoro modules. *Journal of Mathematical Physics*.
- Io, I. F., Huang, F. H., & Hsieh, C. T. (2026). Non-Hermitian free-fermion critical systems and logarithmic conformal field theory. *arXiv preprint arXiv:2602.02649*.